David Phan

hungthui27@outlook.com davphan.github.io/Portfolio/ www.linkedin.com/in/dphan3/
github.com/davphan

Education

University of Washington, Seattle Sept. 2020 - June 2024
Bachelor of Science in Electrical and Computer Engineering (ECE) GPA: 3.87, Cum Laude

Professional Experience

Software Development Engineer II - Amazon Jul. 2026 - Present

- Redesigning the authentication model for Amazon's internal media ingestion platform.
- Enforcing automated security guardrails on ingestion, detection, and transformation of media content at the scale of 1B+ workflows a day.
- Leading the team by organizing and coordinating task distribution across multiple projects, and mentoring interns to guide their development.
- Modernizing legacy systems by migrating to OpenTelemetry for tracing and analytics data across distributed systems.

Software Development Engineer I - Amazon Sept. 2024 - Jul. 2026

- Operated Amazon's core Java/Guice based distributed media ingestion REST API service, managing operational tasks such as oncall rotations and peak event preparation.
- Modernized Amazon's content ingestion by adding support for media types such as HEIF and rebuilding components in Rust for lower latency and memory efficient file type operations, reducing monthly incompatibility ingestion related errors by 32%.
- Enhanced team operations effectiveness by leading monthly bug bash sessions, onboarding new team members to our team oncall rotations, and migrating our team to new task management tooling.
- Migrated legacy bare-metal deployment infrastructure to CDK based AWS cloud infrastructure.
- Recognized as a top 5 leader within my org in pushing Amazon's leadership principles, leading to an accelerated promotion timeline.

Software Development Engineer Intern - Amazon Jun. 2023 - Sept. 2023

- Spearheaded internal full-stack application project to aid ML research teams in processing datasets.
- Designed and implemented both a React/Typescript front-end and a REST API using AWS API Gateway, Lambda, s3, DynamoDB, and Amazon internal tools.
- Deployed applications through CI/CD frameworks using Git, internal pipeline tools, and AWS Pipelines.
- Collaborated with a team of developers to generate and meet design goals using the Agile Methodology/Framework.

Software Engineering Intern - nFlux.ai Sept. 2021 – Dec. 2021

- Analyzed and processed video files for ML training and worked with senior engineers on DevOps tasks to develop CI/CD pipelines using Docker Compose and Git Actions to enhance project workflow.
- Designed and developed python code that automated and organized database information retrieval for SQL and noSQL databases.
- Developed functionality testing procedures to analyze UI/UX components of company products.

Research Experience

Software Development Intern - Makeability Lab, Project Sidewalk Mar. 2022 – Jun. 2024

- Maintain and update the Project Sidewalk API and website, which tracks and updates geographical information on pedestrian accessibility issues in current city infrastructure.
- Collaborate with a team of software developers to respond to GitHub tickets to quickly debug issues and

implement new features in the code base or frontend UI/UX.

- Utilize technologies like Scala, PostgreSQL, and other frontend languages/frameworks.

Undergraduate Researcher - Theberge (BCME) Lab Jul. 2022 – Jun. 2024

- Automated data-analytics and image processing tasks by coding python-based desktop applications.
- Designed and prototyped microfluidic and biomedical devices for research and medical purposes using CAD software, 3D printing and Milling devices, and microcontroller circuits.
- Built and optimized website features to represent lab research and expand study group participation.

Teaching and Mentoring Experience

Undergraduate Electrical and Computer Engineering Tutor Jan. 2024 - Jun. 2024
UW College of Engineering

- Guided students through 11 different introductory and high level ECE coursework (2xx-4xx level courses) employing combinations of teaching strategies such as Dual Coding (vocal and visual teaching).

Engineering Embassador Mar. 2022 - Jun. 2023
UW College of Engineering

- Led future students towards a STEM pathway by guiding them towards different areas of interest, creating engaging presentations and activities.
- Coordinated events by working in small teams and reporting out in larger conference meetings.
- Analyzed demographic data and statistics in engineering to understand and act on diversity issues.

Publications

On the Dynamic Contact Angle of Capillary-Driven Microflows in Open Channels Mar. 2024
Tokihiro, J.C., McManamen, A.M., **Phan, D.N.**, Thongpang, S., Blake, T.D., Theberge, A.B., & Berthier, J.

- Published in *Langmuir*, 40(13), 7215–7224. DOI: 10.1021/acs.langmuir.4c00391

Research In Your Mailbox: Remote Blood Self-sampling Enables Participation of Underserved Populations in Longitudinal Studies Feb. 2026
Stefanovic, F., Robertson, I., Moloney, K., Edelson, J., Nguyen, S., Shinkawa, V., Uchimura, K., Lin, A., Le, L., Tokihiro, J.C., Takezawa, M.G., **Phan, D.**, Schiffer, J.T., Boeckh, M., Adams, K.N., Waghmare, A., Errett, N.A., Berthier, E., Lim, F.Y., & Theberge, A.B.

- Preprint, *medRxiv*. DOI: 10.64898/2026.02.05.26345688

Other Experience

HackUPC Hackathon May 2021

- Developed front end webpage for a facial recognition security system using HTML, CSS, and the Bootstrap 4 framework.

UW Hyperloop Propulsions Team Oct. 2020 – Jul. 2021

- Developed coil-windings and prototype CAD models for a Dual-Sided Linear Induction Motor (DSLIM), utilizing independent self-guided research and team communication to design UW's first DSLIM.
- Led planning and communications for manufacturing outsourcing of material components.

Washington Aerospace Scholars (WAS) Nov. 2017 – Aug. 2019

- Designed a simulated manned Mars mission using skills from mission design to fundamental engineering practices learned through a rigorous multi-year long aerospace engineering curriculum.
- Led project management between sub-teams as the propulsions team's Communications Expert and won the William E. Boeing award for leadership.

Pharmacy Technician-in-Training – Walgreens Jun. 2021 – Sept. 2021

- Performed pharmacy technician duties and fostered customer relations behind the counter by problem solving

issues and showing exceptional customer service.

Technical Skills

Programming languages/software: Java, Python, Bash/Command Line scripting, C, C++, Jupyter Notebook, Scala, VS Code, HTML, CSS, Javascript/Typescript, React, Bootstrap 4, Dagger DI, Mockito
Version control/Workflow optimization software: Git, Github, Git Actions, Docker Compose
Database and Cloud service software: AWS, Azure, MongoDB, PostgreSQL
Leadership/Project Management: Engineering Ambassador at UW, ECE Tutor at UW, Officer for Music for Charity at UW, School Music Director, Communications Lead at Washington Aerospace Scholars (WAS)

References

Shilpa Gupta, SDE Manager
in linkedin.com/in/shilpagupta2790

Ashleigh Theberge, Professor
in linkedin.com/in/ashleigh-theberge-69b46923

Jon Froehlich, Professor
in linkedin.com/in/jonfroehlich

Michael Saugstad, Software Developer & Data Scientist (Mentor)
in linkedin.com/in/michaelsaugstad